

Practice Quiz 04B — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 04

Counterexamples and correct set identities.

1. (a) Disprove the identity

$$(A \cup B) \setminus C = A \cup (B \setminus C)$$

by giving finite sets A, B, C and computing both sides.

Solution Take $A = C = \{0\}$ and $B = \emptyset$. Then

$$(A \cup B) \setminus C = \{0\} \setminus \{0\} = \emptyset,$$

whereas

$$A \cup (B \setminus C) = \{0\} \cup \emptyset = \{0\}.$$

The two sides differ, so the proposed identity is false. ■

- (b) Express $(A \cup B) \setminus C$ correctly as a union of two set differences. Prove your identity by a chain of membership equivalences, naming the definitions and logical law used.

Solution The correct identity is

$$(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C).$$

For an arbitrary object x , the definitions of union and difference give

$$x \in (A \cup B) \setminus C \iff ((x \in A) \vee (x \in B)) \wedge (x \notin C).$$

By distributivity this is equivalent to

$$((x \in A) \wedge (x \notin C)) \vee ((x \in B) \wedge (x \notin C)).$$

By the definitions of difference and union, this in turn is equivalent to $x \in (A \setminus C) \cup (B \setminus C)$. Hence the two sets are equal. ■